



Ultra-low-capacitance probes have also been introduced with wide bandwidths at probe tips, while the size of the probe tip has been reduced to avoid shorting between components in densely-packed circuit boards. Thus, these provide signals to the acquisition system correctly.

Measuring SI with VNA using frequency multipliers

Due to the demand for higher-data-rate devices, signal integrity (SI) test has become more important nowadays. Earlier, vector network analysers (VNAs) were only used to measure 'S' parameters (input-output relationship between terminals in an electrical network) in telecom instruments, but now these have been upgraded to cover signal integrity also. These VNAs work in frequency domain and have now been upgraded to 120GHz.

VNAs are used for micro-level applications like PCB analysis, where these can point out problems like crosstalk, track discontinuity and copper trace width.

There may also be instances of signal fluctuations, called jitters, which mainly occur in clock circuitry but can also arise from power supply noise, crosstalk or even phase-locked loop circuits.

Madhukar Tripathi, senior manager, marketing and channel sales, Anritsu India, says that to increase the frequency bandwidth of vector network analysers, external frequency multipliers are added as per the application requirements, like an antenna that needs to operate beyond 100GHz.

Thus, testing of high-speed digital circuits or high-bandwidth devices becomes easy for designers, saving their precious time to innovate for future. Tripathy notes, "VNA upgrade has brought about just 20-40 per cent increase in existing prices."

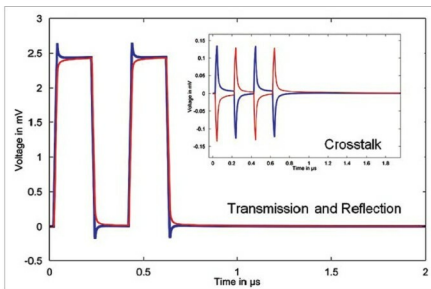


Fig. 2: Image illustrating the crosstalk effect

Standby power measurement using power analysers

Power integrity device players too have introduced upgrades, though at a small scale, to measure voltage and current parameters responsible for power integrity.

Traditionally, power analysers have been used to measure only the input parameters of a line property, when connected on the input side. However, now these have been upgraded to measure power integrity as well. These are placed on the output side of the product to measure output parameters and check power integrity.

Nitin Nigam, application engineer, Tektronix, tells that the bandwidth of power analysers has been increased to use them for PI measurement and also meet application requirements. He adds that power analysers are now coming with an additional functionality of standby power mode.

This has been introduced following the government's directive to save power when devices are not in use. Like mobile chargers, all systems consume a small amount of power even when there is no load but the mains supply is 'on.' A power analyser as a PI instru-

ment measures the system's power consumption in standby mode. So, products that are tested for power integrity, give consumers the benefit of power saving. Monetarily, it is a huge benefit to the manufacturers as well.

In power integrity measurement, oscilloscopes with limited bandwidth are preferred as these produce less noise. If a measurement does not require the bandwidth to be extended, the bandwidth is limited to reduce the noise produced.

The way forward

Earlier, designers used post-layout analysis, which didn't guarantee design success. But now by using high-speed design, analysis and verification techniques early in the design phase, designers can eliminate layout iterations and deliver products on time. As SI and PI issues will continue to remain a big concern, the measurement market will evolve and merge to make it easier for designers. But, at present, a more practical approach to measure power integrity and signal integrity is to have instruments that measure them separately. **EFY**